1.04 Graph Sketching: Linear Graphs

Worked Solutions

1 a)

$x = 3$

1 b)

$y = -4$

2 a)

$m = \frac{6-0}{0-(-2)} = 3$

$y - 6 = 3(x - 0) \Rightarrow y - 6 = 3x \Rightarrow -3x + y = 6$

2 b)

$m = \frac{1-(-1)}{-5-0} = -\frac{2}{5}$

$y + 1 = -\frac{2}{5}(x - 0) \Rightarrow 5y + 5 = -2x$

$\Rightarrow 2x + 5y = -5$

2 c)

$m = \frac{\frac{3}{2}-(-\frac{5}{2})}{2-0} = 2$

$y + \frac{5}{2} = 2(x - 0) \Rightarrow y + \frac{5}{2} = 2x \Rightarrow 2y + 5 = 4x$

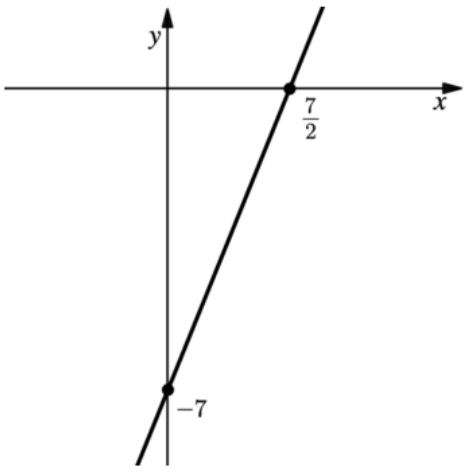
$\Rightarrow -4x + 2y = -5$

2 d)

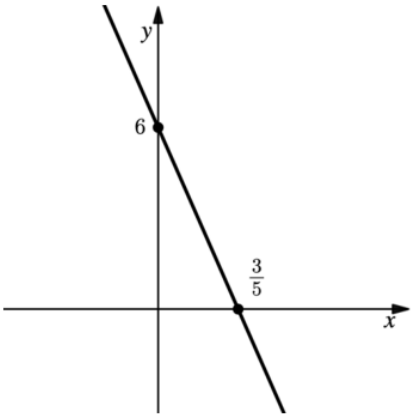
$m = \frac{4-5}{3-0} = -\frac{1}{3}$

$y - 5 = -\frac{1}{3}(x - 0) \Rightarrow 3y - 15 = -x \Rightarrow x + 3y = 15$

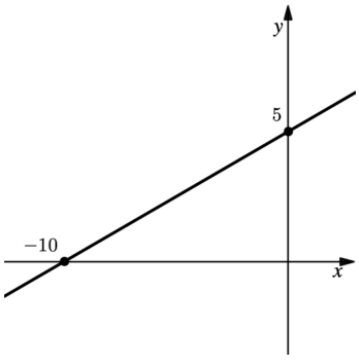
3 a)



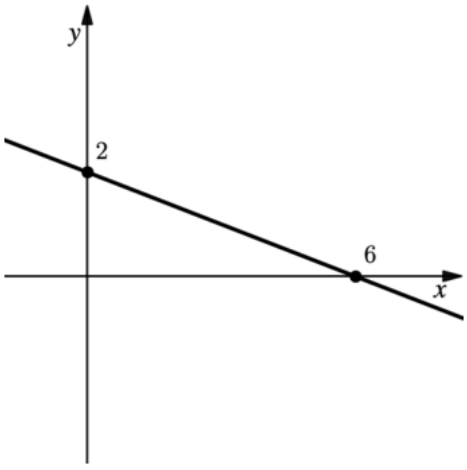
3 b)



3 c)



3 d)



4 a)

$m = \frac{-\frac{3}{4}-\frac{9}{4}}{\frac{1}{2}-(-1)} = -2$

$y - \frac{9}{4} = -2(x+1) \Rightarrow y - \frac{9}{4} = -2x - 2$

$\Rightarrow 4y - 9 = -8x - 8 \Rightarrow 8x + 4y = 1$

4 b)

$m = \frac{\frac{16}{5}-(-\frac{9}{5})}{\frac{4}{5}-(-\frac{6}{5})} = \frac{5}{2}$

$y - \frac{16}{5} = \frac{5}{2}(x - \frac{4}{5}) \Rightarrow y - \frac{16}{5} = \frac{5}{2}x - 2$

$\Rightarrow 2y - \frac{32}{5} = 5x - 4 \Rightarrow 10y - 32 = 25x - 20$

$\Rightarrow -25x + 10y = 12$

4 c)

$m = \frac{\frac{27}{8}-\frac{37}{40}}{1-\frac{3}{10}} = \frac{7}{2}$

$y - \frac{27}{8} = \frac{7}{2}(x-1) \Rightarrow y - \frac{27}{8} = \frac{7}{2}x - \frac{7}{2}$

$\Rightarrow 8y - 27 = 28x - 28 \Rightarrow -28x + 8y = -1$

4 d)

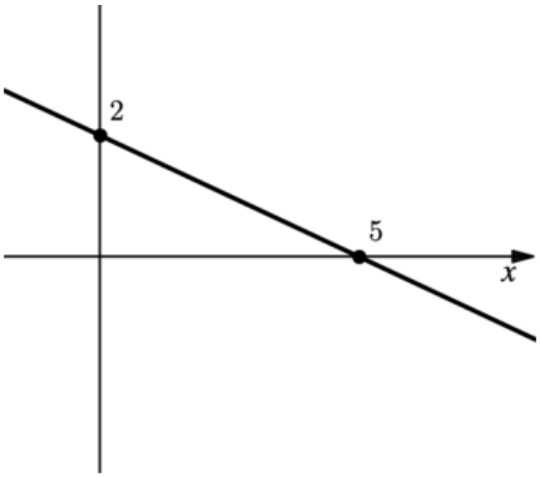
$m = \frac{-\frac{47}{20}+\frac{67}{40}}{3-\frac{3}{10}} = -\frac{1}{4}$

$y + \frac{67}{40} = -\frac{1}{4}(x - \frac{3}{10}) \Rightarrow y + \frac{67}{40} = -\frac{1}{4}x + \frac{3}{40}$

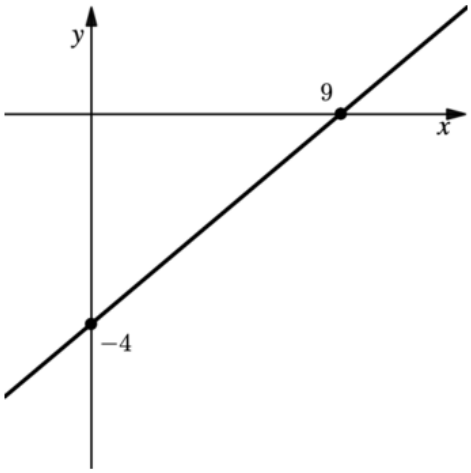
$\Rightarrow 40y + 67 = -10x + 3 \Rightarrow 10x + 40y = -64$

$\Rightarrow 5x + 20y = -32$

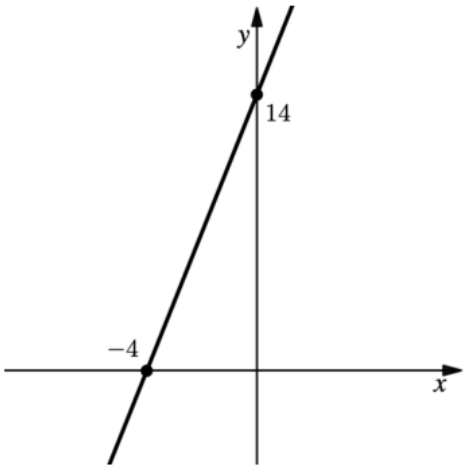
5 a)



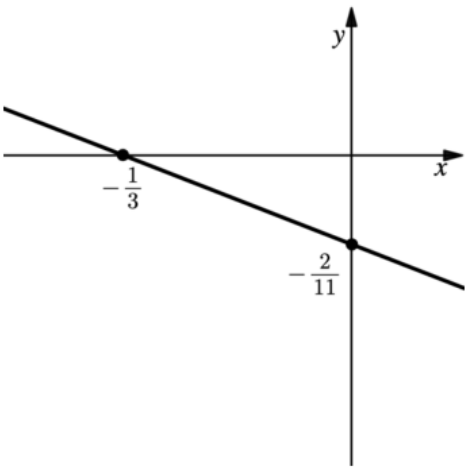
5 b)



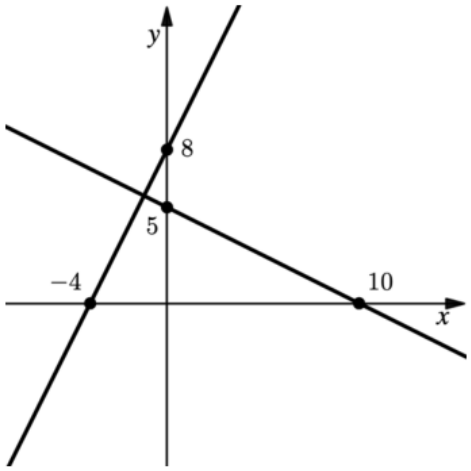
5 c)



5 d)



6 a)



6 b)

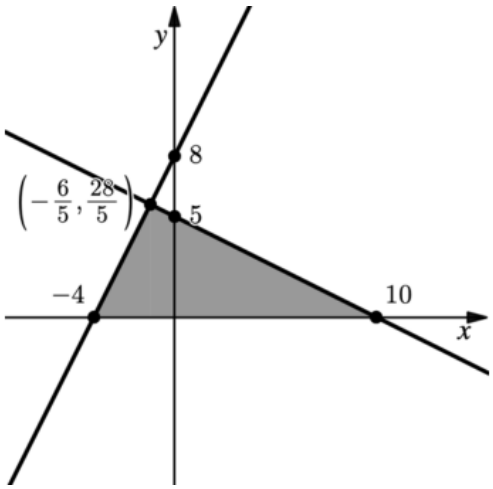
$2(2x + 8) = 10 - x \Rightarrow 4x + 16 = 10 - x$

$\Rightarrow 5x = -6 \Rightarrow x = -\frac{6}{5}$

$\therefore y = 2 \times -\frac{6}{5} + 8 = \frac{28}{5}$

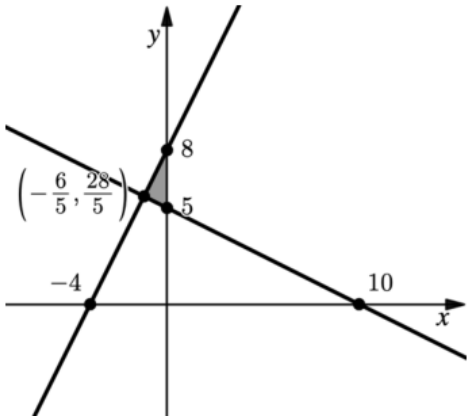
So the lines intersect at $(-\frac{6}{5}, \frac{28}{5})$

6 c)



Area $= \frac{1}{2} \times (10 + 4) \times \frac{28}{5} = \frac{196}{5}$

6 d)



Area $= \frac{1}{2} \times (8-5) \times \frac{6}{5} = \frac{9}{5}$

7 a)

$m = \frac{-\frac{6}{5}-\frac{8}{3}}{-\frac{21}{5}+\frac{61}{9}} = -\frac{3}{2}$

$y - \frac{8}{3} = -\frac{3}{2}(x + \frac{61}{9}) \Rightarrow -\frac{2}{3}y + \frac{16}{9} = x + \frac{61}{9}$

$\Rightarrow x + \frac{2}{3}y = -5$

7 b)

$m = \frac{\frac{9}{2}+\frac{2}{3}}{\frac{67}{30}-\frac{6}{5}} = 5$

$y - \frac{9}{2} = 5(x - \frac{67}{30}) \Rightarrow \frac{1}{5}y - \frac{9}{10} = x - \frac{67}{30} \Rightarrow x - \frac{1}{5}y = \frac{4}{3}$

7 c)

$m = \frac{-\frac{9}{2}-\frac{8}{3}}{\frac{283}{80}+\frac{41}{15}} = -\frac{8}{7}$

$y - \frac{8}{3} = -\frac{8}{7}(x + \frac{41}{15}) \Rightarrow -\frac{7}{8}y + \frac{7}{3} = x + \frac{41}{15}$

$\Rightarrow x + \frac{7}{8}y = -\frac{2}{5}$

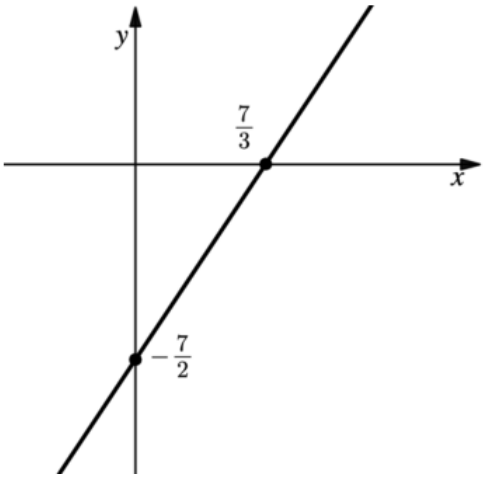
7 d)

$m = \frac{\frac{27}{50}+\frac{6}{5}}{\frac{193}{30}+\frac{31}{6}} = \frac{3}{20}$

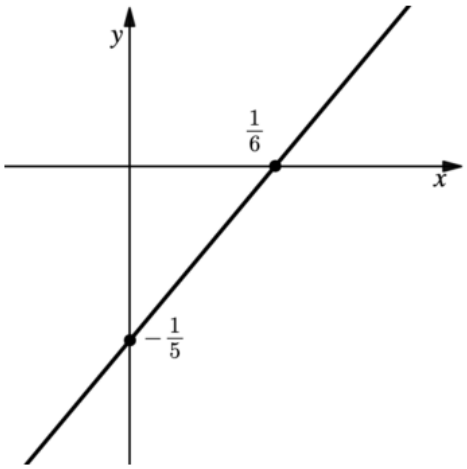
$y + \frac{6}{5} = \frac{3}{20}(x + \frac{31}{6}) \Rightarrow \frac{20}{3}y + 8 = x + \frac{31}{6}$

$\Rightarrow x - \frac{20}{3}y = \frac{17}{6}$

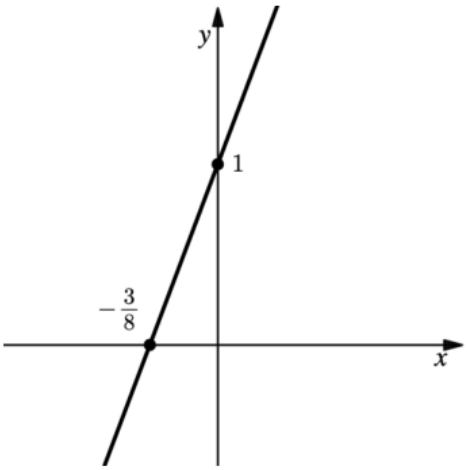
8 a)



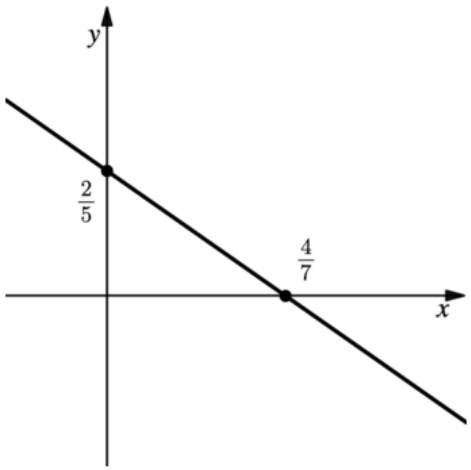
8 b)



8 c)



8 d)



9)

$m = \frac{(3k-1)-(k+1)}{(2k)-(-k)} = \frac{2k-2}{3k}$

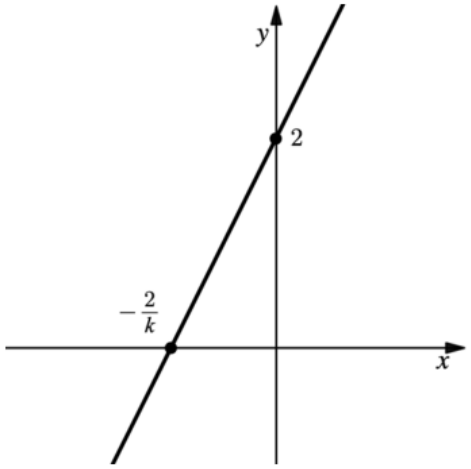
$y - (k+1) = \frac{2k-2}{3k}(x+k)$

$\Rightarrow y - (k+1) = \frac{2k-2}{3k}x + \frac{2k-2}{3}$

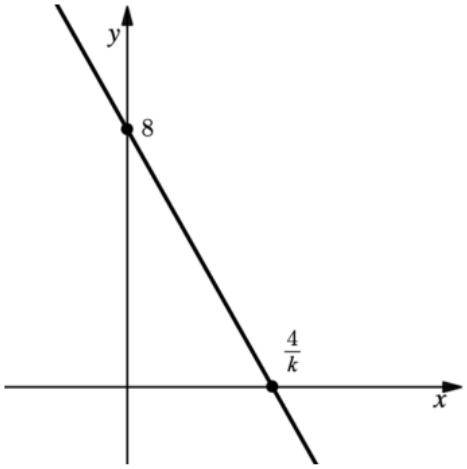
$\Rightarrow y = \frac{2k-2}{3k}x + \frac{2}{3}k - \frac{2}{3} + k + 1$

$\Rightarrow y = \frac{2k-2}{3k}x + \frac{5}{3}k + \frac{1}{3} \Rightarrow y = \frac{2k-2}{3k}x + \frac{5k+1}{3}$

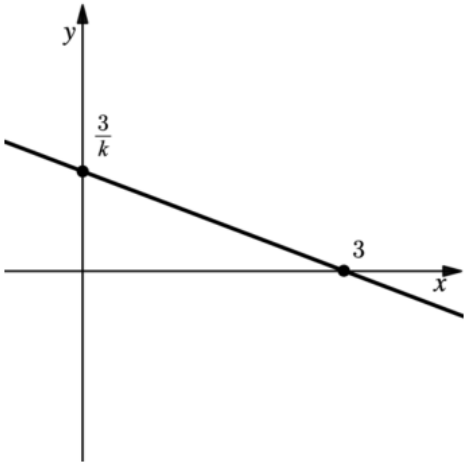
10 a)



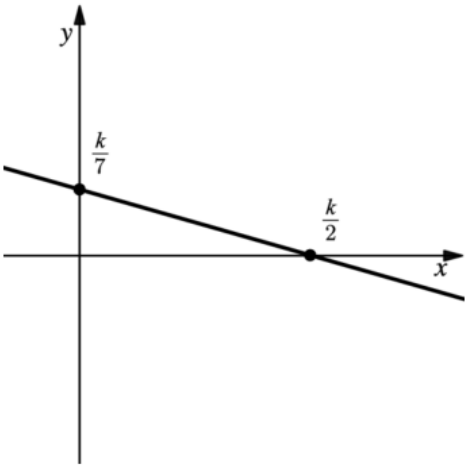
10 b)



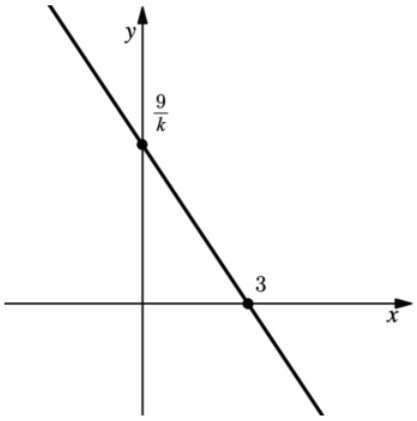
10 c)



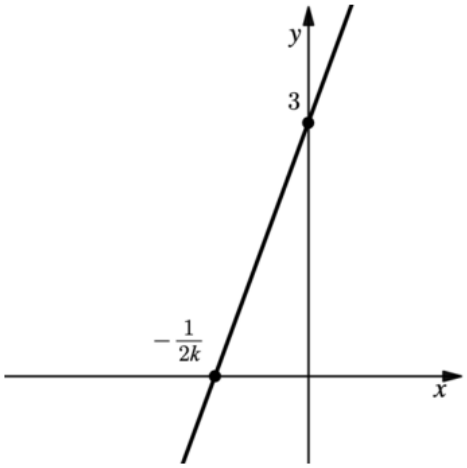
10 d)



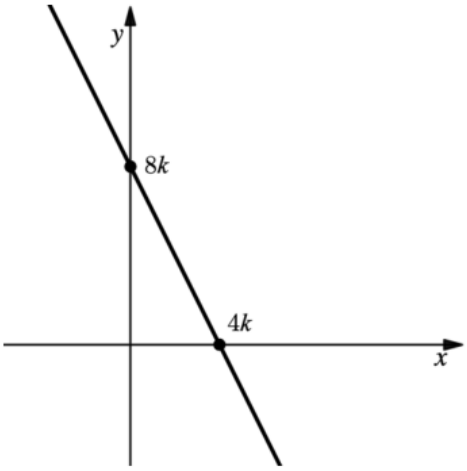
10 e)



10 f)



10 g)



11 a)

$m = \frac{(12k^2-5k)-(-3k^2-5k)}{4+1} = \frac{15k^2}{5} = 3k^2$

$l_1 : y - (12k^2 - 5k) = 3k^2(x - 4)$

11 b)

$m = \frac{(\frac{3}{k}+\frac{1}{k^2})-(\frac{3}{k}-\frac{5}{k^2})}{-2-10} = \frac{\frac{6}{k^2}}{-12} = -\frac{1}{2k^2}$

$l_2 : y - (\frac{3}{k} + \frac{1}{k^2}) = -\frac{1}{2k^2}(x + 2)$

11 c)

$l_1 - l_2$:

$$(y - (12k^2 - 5k)) - (y - (\frac{3}{k} + \frac{1}{k^2})) = (3k^2(x - 4)) - (-\frac{1}{2k^2}(x + 2))$$
$$\Rightarrow -12k^2 + 5k + \frac{3}{k} + \frac{1}{k^2} = 3k^2x - 12k^2 + \frac{1}{2k^2}x + \frac{1}{k^2}$$
$$\Rightarrow 5k + \frac{3}{k} = (3k^2 + \frac{1}{2k^2})x$$
$$\Rightarrow x = \frac{5k + \frac{3}{k}}{3k^2 + \frac{1}{2k^2}} = \frac{10k^3 + 6k}{6k^4 + 1}$$

Then:

$$y - (12k^2 - 5k) = 3k^2(\frac{10k^3 + 6k}{6k^4 + 1} - 4)$$
$$\Rightarrow y = \frac{30k^5 + 18k^3}{6k^4 + 1} - 5k$$
$$\Rightarrow y = \frac{30k^5 + 18k^3}{6k^4 + 1} - \frac{5k(6k^4 + 1)}{6k^4 + 1}$$
$$\Rightarrow y = \frac{30k^5 + 18k^3}{6k^4 + 1} - \frac{30k^5 + 5k}{6k^4 + 1}$$
$$\Rightarrow y = \frac{18k^3 - 5k}{6k^4 + 1}$$

So l_1 and l_2 intersect at $(\frac{10k^3+6k}{6k^4+1}, \frac{18k^3-5k}{6k^4+1})$.